Boundary Shape-Preserving Model for Building Mapping From High-Resolution Remote Sensing Images

Anna Hu, Liang Wu, Siqiong Chen, Yongyang Xu, Haitao Wang, and Zhong Xie

***Abstract*— The building is a critical urban element for developing smart cities. Building extraction research has experienced rapid development due to abundant remote sensing image data availability. Building extraction methods have recently focused on enhancing regional accuracy without obtaining refined building boundaries and do not fit the concave–convex shape of the building and have significant sawtooth noise. This article proposes a building shape-preserving framework to solve the imperfect boundary problem. To delineate the concave and convex building boundaries from remote sensing images, we combine coarse- and fine-grained information with the instance segmentation method, Mask region-based convolutional neural network (R-CNN), which can further refine the indeterminate boundary points, and perform inward fitting of the building boundary. A regular boundary network is designed to learn the features of the edges and footprints to calculate the boundary loss that can smoothen the boundary noise, considering the imperfect boundaries and sawtooth noise in the extracted building boundary. Furthermore, to enhance the ability to evaluate the extracted building shape, a building footprint information evaluation algorithm, the footprint distance metric, is proposed to calculate the footprint distance between each building boundary. Comparing the building extraction results indicates that the proposed method achieves excellent quantitative evaluation scores. Experiments performed using Spacenet and Wuhan University (WHU) datasets to verify the effectiveness of the proposed method are detailed. The experimental results indicate that the algorithm utilizes the powerful edge extraction capability of the method to exhibit excellent performance in terms of extracting building edges from complex remote sensing scenes. Code is available at https://github.com/AnnaCUG/Boundary-shape-preserving-model.**

Manuscript received 21 September 2022; revised 13 December 2022, 10 April 2023, and 8 May 2023; accepted 11 May 2023. Date of publication 18 May 2023; date of current version 1 June 2023. This work was supported in part by the National Natural Science Foundation of China under Grant 42001340 and Grant 41871311, in part by the State Key Laboratory of Geo-Information Engineering under Grant SKLGIE2021-M-4-1, in part by the Guangdong Science and Technology Strategic Innovation Fund (Guangdong-Hong Kong-Macau Joint Laboratory Program) under Grant 2020B1212030009, and in part by the Open Research Project of the Hubei Key Laboratory of Intelligent Geo-Information Processing under Grant KLIGIP-2022-A05. *(Corresponding author: Yongyang Xu.)*

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Digital Object Identifier 10.1109/TGRS.2023.3277699

***Index Terms*— Boundary shape-preserving, building extracted, building mapping, instance segmentation, remote sensing imagery.**

I. Introduction

With the rapid development of remote sensing technology, high-resolution optical remote sensing images have become the leading data resource for efficient ground object extraction. The successful application of deep learning in computer vision has encouraged other data-driven research, such as remote sensing image processing [1], [2]. To develop more intelligent remote sensing image processing methods, researchers and experts introduced deep learning methods to extract ground objects [3] and perform terrain classification [4] and scene recognition [5]. Unlike conventional machine learning methods, the performance of deep learning methods can constantly improve owing to its outstanding ability of data feature extraction [6], [7].

Initially, researchers directly introduced semantic segmentation methods in deep learning to extract building contour information from remote sensing images [8], [9]. These semantic segmentation methods generate binary building contour information through end-to-end learning of visible light remote sensing images. Because of the complexity of the terrain environment in remote sensing images, simple feature extraction methods cannot accurately distinguish buildings from other terrain objects. Researchers have also designed feature extraction methods for building textures to improve the extraction accuracy [10], [11], [12], [13]. Because a building is an artificial ground object with independent characteristics, the extraction task for a single building is essential. Therefore, researchers have introduced an instance segmentation method to extract individual buildings. Recent studies on building extraction have also proposed a histogram thresholding Mask region-based convolutional neural network (R-CNN) [14], which exhibited outstanding performance in the case of limited training samples [15].

Utilizing the benefits of feature extraction in deep learning methods, the current building extraction methods have shown excellent performance in terms of accuracy. However, most methods aim to enhance the textural features of the interior of the building, and the shape information is ignored. Owing to the similarity in the spectral information of ground objects in remote sensing images, problems of irregular and unsmooth boundaries still exist in recent extraction results. As artificial objects, buildings have regular, smooth shape features. The building boundary extraction methods must consider the

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shortcomings of building outlines, which involve irregular building boundaries (such as irregular right angle corners).

Several studies have been conducted to solve the problem of incomplete building boundary results from high-resolution optical remote sensing images. These studies fully utilized the textural and spectral information to distinguish a building from other ground objects [16], [17]. These methods use the active contour [17], [18], [19] and recurrent neural network (RNN) methods [20]. The active contour methods utilize an encoder–decoder architecture and a neural renderer to create a polygon from its vertices. The RNN extraction methods, which use RNN, long short-term memory (LSTM), and other improved LSTM blocks to locate footprints, provide regular building boundaries.

Although these instance segmentation methods distinguish each building in remote sensing images, vector footprints can be obtained by performing vector deletion conversion from a binary image. In addition to the limited ability to extract vector boundary footprints, this method focuses on solving the problems of boundary optimization, which can also be migrated to vector methods. Recent building extraction methods must also optimize the extraction results using the shape of the building boundary in remote sensing images. Mainstream research on building extraction adopted various methods to improve the extract accuracy, such as enhancing the texture feature, extracting building location information, and combining contextual features [12]. But, those methods ignore the boundary shape problem and do not use boundary and footprint information. Furthermore, the building shape exhibits a sawtooth phenomenon in the extracted boundary results. The building boundary is smooth in the scene, and a single boundary comprises two footprints. However, recent extraction results have several footprints, and these points cannot be composed as a single boundary. Finally, the widely used intersection over union (IoU) [21] shape similarity measurement method is not sensitive to the building shape, such as the footprints and visual perception.

To solve the abovementioned challenges, we reconsidered the effectiveness of extracting boundary information and proposed several methods for boundary feature enhancement. The following are the key contributions of this study.

1) A shape preservation model (SPM) was proposed, and the original mask-generated branch in Mask R-CNN was replaced. This network was devoted to solving the problem of unsmooth building boundaries and enlarging the current building boundary shape information.
2) The regular boundary model provides boundary and footprint information to determine the regular building shape. Subsequently, the proposed boundary loss was used to optimize the boundary and footprint errors using the Euclidean space distance.
3) To ensure a more objective assessment of the extracted building mask, this study applied both the conventional IoU score and the polygons and line segment (PoLiS) metrics for evaluation, which was improved using the PoLiS method [22]. This approach is robust to different vertices and sensitive to slight variations in translation, rotation, and scale.

II. Related Work

Several researchers have performed building extraction from remote sensing images. The methods used in these studies can be classified into morphological, conventional machine learning, and deep learning methods. In particular, deep learning methods have been significantly improved, and it can also be classified into semantic segmentation, instance segmentation, and boundary optimized methods.

A. Morphological Building Extraction Methods

As the pioneering building extraction method, morphological methods generally use a single feature of the buildings in a high-space resolution remote sensing image. At this stage, researchers use additional information to facilitate the extraction of building boundaries [23], [24], [25]; altitude and shadow information was used to distinguish the building from the road and other ground objects. These studies added digital surface model (DSM) data and DSM slope information to facilitate the building extraction feature. Moreover, the index model is also used in remote sensing image building extraction. Index model methods can decrease the typical training steps of samples and segment high-resolution remote sensing images. The method [26], [27] uses a morphological approach to identify a single characteristic of a building and determine this characteristic from a complex remote sensing scene. Other researchers have focused on extracting building objects, not a single pixel, that can fully use the semantic information. These methods [28], [29] use machine learning techniques, such as support vector machine (SVM), to classify the regional characteristics. Irrespective of what these methods focus on while describing the unique characteristics of buildings, they identify single features of a building but cannot capture the overall detailed information.

B. Deep Learning Building Extraction Methods

To solve the problems of the existing morphological methods and conventional machine learning methods, deep learning methods have been proposed to extract critical geographical elements in remote sensing images, such as roads [30], [31] and buildings [32], [33]. Because of their significant capability to learn complex representations of building structures from large volumes of remote sensing image, deep learning methods exhibit outstanding performance in extraction accuracy. Moreover, these deep learning methods can overcome the effects of a complex remote sensing scene and facilitate the extraction of geographical elements [34], [35]. Owing to the benefits of semantic segmentation methods [36], [37] in computing vision, research on building extraction has extensively developed in recent years. However, pixel-level methods face a severe problem in building boundary extraction in that the semantic segmentation results cannot distinguish the position of each building. To overcome this problem, the instance segmentation method has been proposed [34], [35]. The instance segmentation method is a broader technique for extracting building boundaries and can distinguish each building from the original building category.

Several outstanding instance segmentation models have been established, such as the Mask R-CNN, tensor mask [38],

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and blend mask [39]. Several studies have been conducted to develop the appropriate object extraction networks, and they have achieved a high IoU accuracy. Instance segmentation methods have been extensively researched [35], [40], [41], [42]. Recently, object-oriented methods have focused on solving the problem of building boundaries. Some researchers have attempted to solve this problem and focus on the morphological approach to capture the building boundary information.

C. Boundary Optimized Methods

Several researchers are aware of building boundary problems, and several boundary optimization methods have been proposed to solve these problems. The cycle loss function [43], attention block and coarse-to-fine boundary optimization block [44], and vector building outline generation methods [33] were proposed to optimize a building boundary. Although these methods enhance the boundary features, the boundary of the extracted building results still exhibits substantial defects affecting the interpretation applications (such as the building height extract) to a large extent. The boundary optimization methods combine deep learning with a morphological algorithm to fit the boundary details. A study [33] demonstrated an end-to-end method, which used a PointNet block to fit the building boundary points and extract building boundaries from the patch remote sensing images. Footprint extraction methods extract building footprints and connect the extracted footprints to obtain regular boundaries. These methods [20], [45] connect the feature extraction network with RNN, LSTM, and other improved LSTM blocks to find footprints and obtain regular building boundaries. Although these methods focus on the refined extraction of building boundaries, they can only extract the boundaries of one building in each process, which is unsuitable for practical applications. Moreover, these methods use a one-hot strategy representing building footprints, because they extract vector footprints. This strategy is limited to preserving the spatial topological relationships of each footprint and requires considerable computation.

In addition to the difficulties encountered in refining building edge extraction methods, standardizing the evaluation techniques for building extraction is an unresolved issue in remote sensing. IoU is the most popular assessment algorithm used in object extraction methods and has been employed in several studies. This method determines the distance between the predicted object boundary and the original object boundary, which is the ratio of the two objects' intersection and union values. With the development of the object detection method, the IoU has been further strengthened at various distances. Rezatofighi et al. [46] proposed a general object distance assessment method considering that the predicted object boundary is disjointed from the original object boundary. This method solves the problem of the conventional IoU value being 0, and this error cannot be used as feedback for the object extraction method. The generalized IoU (GIoU) method uses a minimum bounding rectangle area to outperform other methods in terms of efficiency and to reflect a typical situation. Moreover, to further add the overlap area, center point distance, and length–width ratio, Zheng et al. [47] proposed the use of the distance IoU (DIoU) and compared IoU (CIoU) and replaced the nonmaximum suppression (NMS) with the DIoU-NMS. The two assessment methods exhibited a faster convergence speed and outstanding results. Although the methods can be used to solve the problems associated with the IoU method to a considerable extent, they only consider pixel differences in the building shape. These three rates are insensitive to additional points on polygons and linearly respond when tiny changes occur in translation, rotation, and scaling. The main disadvantage of the IoU evaluation method is that it must include topology information in the building footprints.

Considering the significance of building boundaries, we designed a novel mask-generated network, an edge point enlargement algorithm, which combines the shape-preserving network with the regular boundary network. The shape-preserving network was designed to enhance the details of the boundary features in the original mask-generated branch in Mask R-CNN. A regular boundary network was used to solve the problem of the sawtooth boundary instead of merely enhancing the mask branch. The boundary loss was used to optimize the boundary extraction feature using the Euclidean space distance. Finally, the footprint distance metric utilizes vertex and boundary distance in the building and reveals the visual perception.

III. Methodology

This study introduces the Mask R-CNN, one of the most classical object-oriented algorithms used for object detection, as the baseline network architecture. This structure generated pixel segmentation results and used the vector-to-grid algorithm to transition the pixel results to vector footprints. Although Mask R-CNN cannot generate vector footprint results, this study focuses on boundary optimization, not the generated boundary footprints. Considering the disadvantage in the boundaries of the current building boundary extraction methods, we designed an edge point enlargement algorithm to fit the potential edge points (Fig. 1). Furthermore, the designed boundary loss was used to measure the footprint and boundary error, which facilitates the building boundary contour shape. The building footprints are the most crucial feature in a building outline that can accurately reflect the building's position. This study designed the footprint distance metric and optimized the building outline evaluation method. The footprint distance metric can be better utilized in building extraction and can process multiobject datasets.

A. Baseline Algorithm

Mask R-CNN is an exemplary object-oriented method based on faster R-CNN and a fully convolutional network (FCN). In addition to the original faster R-CNN structure, Mask R-CNN uses two optimization blocks: the region of interest align (RoIAlign) structure and the mask detected branch. On the one hand, Mask R-CNN replaces RoI pooling with RoIAlign and improves the unprecise pixel recovery method. The RoIAlign algorithm uses bilinear interpolation to compute the misalignment point and restores the relative position to a precise absolute position. On the other hand, Mask R-CNN

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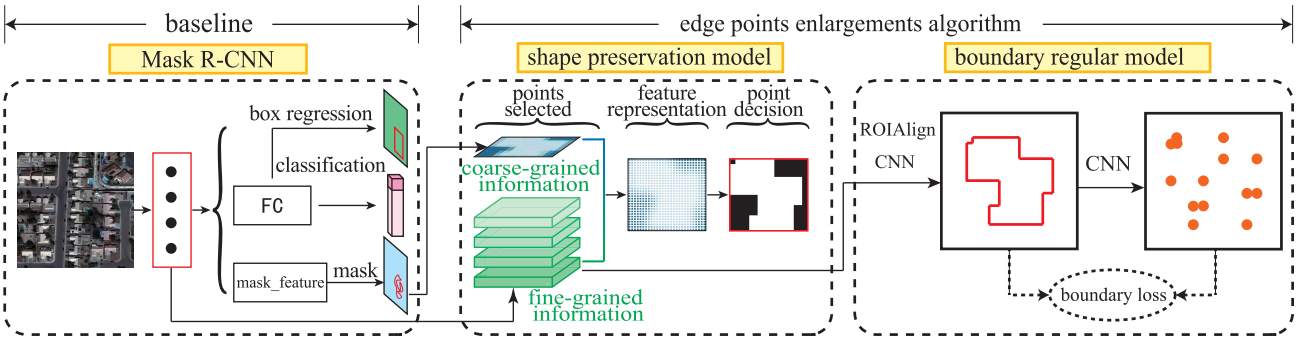


Fig. 1. Structure of the building boundary extract method. This study comprises the baseline, shape preservation, and regular boundary models.

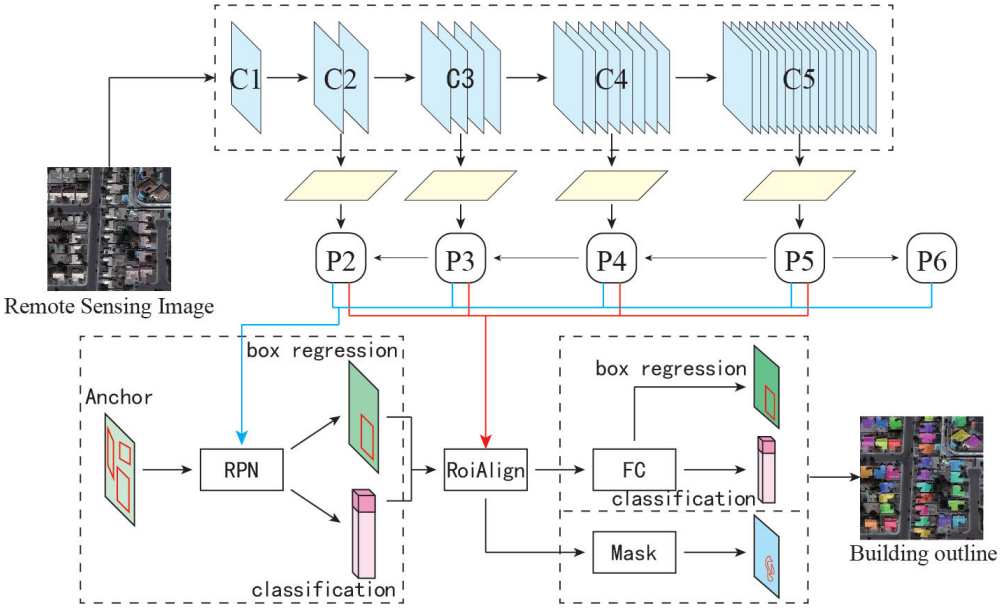


Fig. 2. Structure of Mask R-CNN. *C* and *P* denote the cascading features in the FPN model.

increases the branches for generating segmentation masks on each RoI in parallel with the existing branch for bounding-box regression and classification. The added branch is a small FCN used to predict a segmentation mask pixel-by-pixel for each RoI. Mask R-CNN has been widely applied in ground object extraction from remote sensing images. This study used Mask R-CNN as the baseline method to optimize the boundary problems, which can avoid the influence of other factors (such as performance issues and applicability to remote sensing imaging issues). Although Mask R-CNN is not the most outstanding object extraction method, the stability of this method is the most suitable. The structure of Mask R-CNN is shown in Fig. 2.

For each remote sensing image, Mask R-CNN extracts the object features and multiple dimensional features. The RoI is generated from the feature extraction network using the region proposal network (RPN), which is mapped as the feature vector of the fixed dimension. In the last processing stage, three branches are generated, namely, the object class, regression results of the target box, and instance segmentation mask.

B. Edge Points Enlargement Algorithm

The characteristics of the building boundary are smooth and have regular lines in natural scenes. Semantic segmentation requires that each pixel retains the original shape feature according to the characteristics of different buildings. In general, identifying the points surrounding the edge is challenging, whether in the building or background. The potential boundary points determine the precision of the building boundary. Therefore, boundary thinning for buildings should start with enhancing the ability to classify the potential points.

1) Shape Preservation in Boundary Concave–Convex: The mask branch of Mask R-CNN determines the accuracy of the building boundary. The mask-generation network needs to efficiently handle the problem of building boundaries. The boundary feature is subsampled to a small dimension in the classical semantic segmentation network. However, in Mask R-CNN, the output of the mask-generated network must be superior to improve the model efficiency, which is not conducive to boundary extraction. Although the pixel in the boundary is the easiest misjudgment, the boundary pixel is only a fraction of the total pixels. In this study, the shape-preserving network was designed to address the boundary problem; its structure is shown in Fig. 3. The shape-preserving network is an image segmentation optimization method. In computing vision, the upsample block performs well in the boundary process. The edge point enlargement algorithm combines the upsample and effective pixel segmentation blocks.

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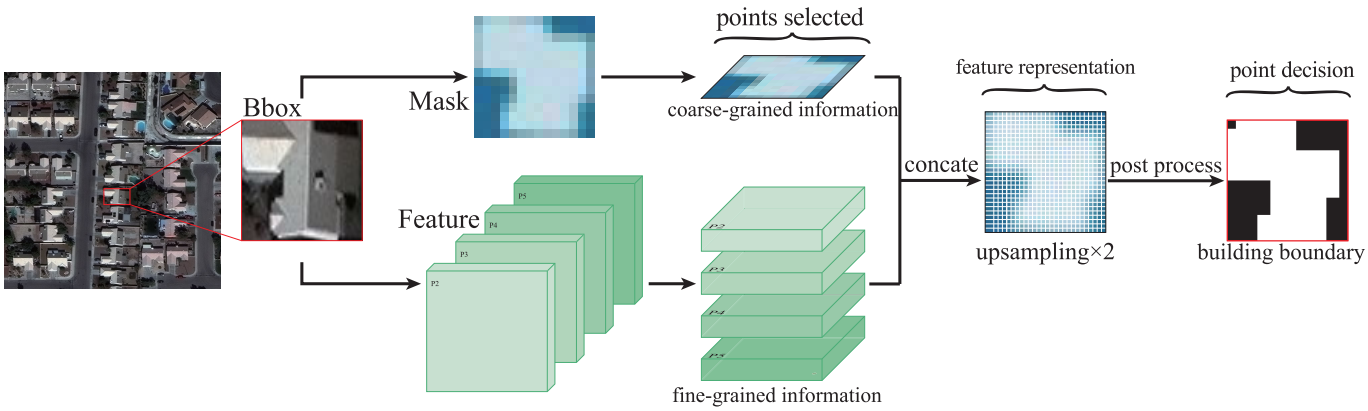


Fig. 3. Structure of the shape-preserving network. The mask is the extraction result of the original Mask R-CNN; the feature is the cascading feature in the FPN model.

In particular, this method fuses fine-grained and coarse-grained information at the beginning of the edge point enlargement algorithm to best use the building boundary feature. Fine-grained information focuses on extracting global context information, which can capture similar building shape information at other regions of the remote sensing image. Similar shape features can facilitate the discrimination of building boundaries. Coarse-grained information focuses on a local feature of the current building outline. Following the details of textural information, the approximate boundary position can be located. By combining fine-grained information with coarse-grained information, the shape preservation block can use rough approximate boundary positions and accurate shapes to solve the problem of concave–convex boundaries. The input of shape-preserving network is combined with the fine-grained and coarse-grained information. The fine-grained information is obtained from the feature pyramid networks (FPNs) [48], and the coarse-grained information is obtained from the original mask feature. The coarse and fine boundary information is concatenated and upsampled to classify building boundary points. The edge point enlargement algorithm contains three main components: a point selection block, a feature representation block, and a point decision block. First, considering the unfriendly gradient descent of the iterative subdivision algorithm in the point selected block, the training and test processes used the irregular and regular point selection strategies, respectively. To select as many potential edge vertices as possible, a nonuniform sampling method was used to obtain boundary points. The number of parameters was significantly increased. Therefore, all pixels in the mask are the potential edge points, which avoids the existing shape error and reduces the number of parameters. Second, in the feature representation block, the feature value of the boundary point comprises fine-grained and coarse-grained information. The upsampling method calculates the probability value of surrounding information; this method can use the subpixel information encoded and obtain higher resolution building mask results. Third, the simple semantic segmentation network discriminates the boundary points in the point decision block.

2) Boundary Regular in Sawtooth Boundary: The most crucial shape factors of the building boundary, footprint, and edge features determine the precision of the building boundary extraction. In this study, a regular boundary network was designed to extract building footprints and boundary features constrained by boundary loss. In the FPN block, the multiscale cascade network combines shallow and deep layer information, yielding positional and high-frequency semantic information. The regular boundary network introduces the multiscale cascade feature to extract building boundary information (boundary feature and points feature) to obtain rich object ground feature information. Because regular boundary loss can be constrained, the model generates a common building boundary. The structure of the regular boundary network is shown in Fig. 4.

The end-to-end building boundary extraction method was used to construct a new building footprint-generated algorithm. This block removes redundant points in the building boundary and determines the key points, which can solve the problem of too many bumpy boundary lines in building boundary extraction. The boundary loss function calculates the pixel error between each building mask and accounts for the edge distance. We added the boundary and footprint features extracted as the final optimized loss function and named the boundary loss. The boundary loss is given as (1). The boundary and point features provide the direction and position information.

C. Overall Loss Functions

In addition to the added segmentation branch, the original Mask R-CNN loss on each sampled RoI is composed of the classification loss (L_{mask}), bounding-box-loss (L_{box}), and segmentation mask loss (L_{cls}). L_{cls} and (L_{box}) are identical to those in Faster R-CNN. L_{mask} is defined as the average binary cross-entropy loss that applies a per-pix sigmoid. An RoI associated with a ground-truth class k depends on the kth mask.

This study designed a boundary loss, which can be expressed as follows:

$$L_{\text{Boundary}} = Y_{\text{pred}}(\text{Dis}(N_{\text{mask}}) \times N_{\text{mask}} - (\text{Dis}(P_{\text{mask}}) - 1) \times P_{\text{mask}}) \quad (1)$$

where Y_{pred} is the extracted building edge or footprints results, P_{mask} is the "bool" value, whereas the "false" value represents the edge or footprints, and the "true" value represents the background in the ground truth, N_{mask} is the inverse value of P_{mask}, and Dis is the exact Euclidean distance transform.

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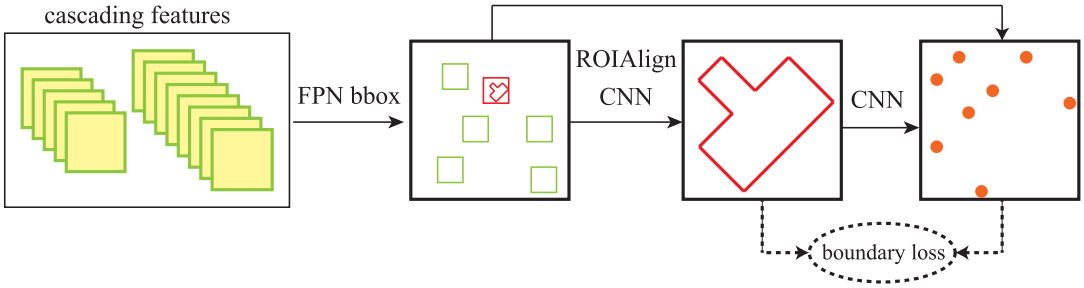


Fig. 4. Structure of the regular boundary network of footprint and edge feature extraction. The cascading features are the outputs of the FPN block.

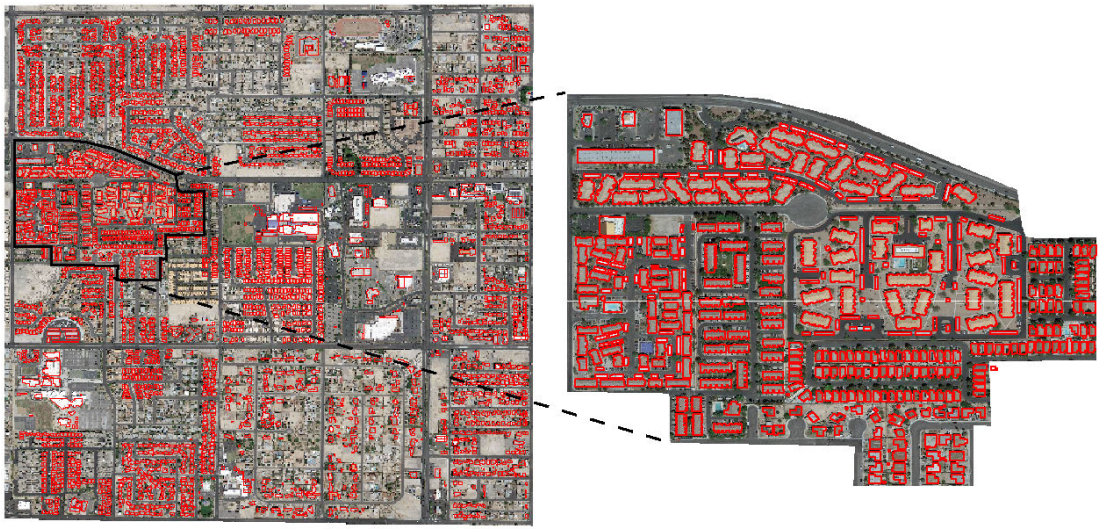


Fig. 5. Example images and the corresponding labels of the Spacenet dataset.

The overall loss function of this building boundary shape-preserving model can be expressed as follows:

$$L = L_{\text{mask}} + L_{\text{box}} + L_{\text{cls}} + L_{\text{Boundary}}. \tag{2}$$

IV. Dataset and Experiment

A. Dataset

To verify the performance and generalization of the algorithm under different scene datasets, two public benchmark datasets, called Spacenet [49] and Wuhan University (WHU) building dataset (WHU) [50], were selected to evaluate the proposed method. The building footprints of the two datasets were stored in annotated files. The format of the footprints is the same as that of the common objects in context (COCO) dataset, which includes coordinating information and object type. Example images and the corresponding labels are shown in Figs. 5 and 6.

1) Spacenet: The Spacenet dataset is a visible light remote sensing image source acquired from the WorldView-3 satellite, with a spatial resolution of 0.3 m. In this study, we used the dataset in Las Vegas, considering the efficiency of the experiment. Each building boundary annotation is a polygon format based on a sequence of vertices, the most commonly used vector data format. In the Spacenet dataset, the training set comprises 3390 images with 104 128 annotated building footprints, and the test set contains 226 images. In the dataset, the size of all the remote sensing images is 650 × 650, which is more significant than those of the datasets in several existing studies. It contains enough building objects in one image. The training and testing datasets contain 3390 and 226 remote sensing images, respectively.

2) WHU: The WHU dataset contains aerial and satellite images covering 1000 km^2, with raster labels and vector maps The image covers most of the building area in the middle of the aerial dataset. To adapt to the size of the network input and reduce hardware computing pressure, a 0.075-m ground resolution remote sensing image was subsampled at a 0.3-m ground resolution, and the original complete image was cut to a size of 512 × 512. Moreover, considering the completion of the building boundary, we solved the problem of cropping the building with holes and multiple parts. While cropping vector building coordinates, the building holes are cut into different objects rather than cropping the holes and buildings into a single building with a concave shape. To solve this problem, we redesigned the data-cropping method, assigned the main body of the building and the holes to different types of objects, and used the shapely[1] package in Python to crop the buildings with holes. Holes and building bodies can be fused during cropping to generate the correct building shape. The training and testing datasets contain 19 221 and 9794 remote sensing images, respectively.

B. Evaluation Metrics

The prediction results of different methods were evaluated and compared using different evaluation criteria.

[1]https://shapely.readthedocs.io/en/stable/installation.html

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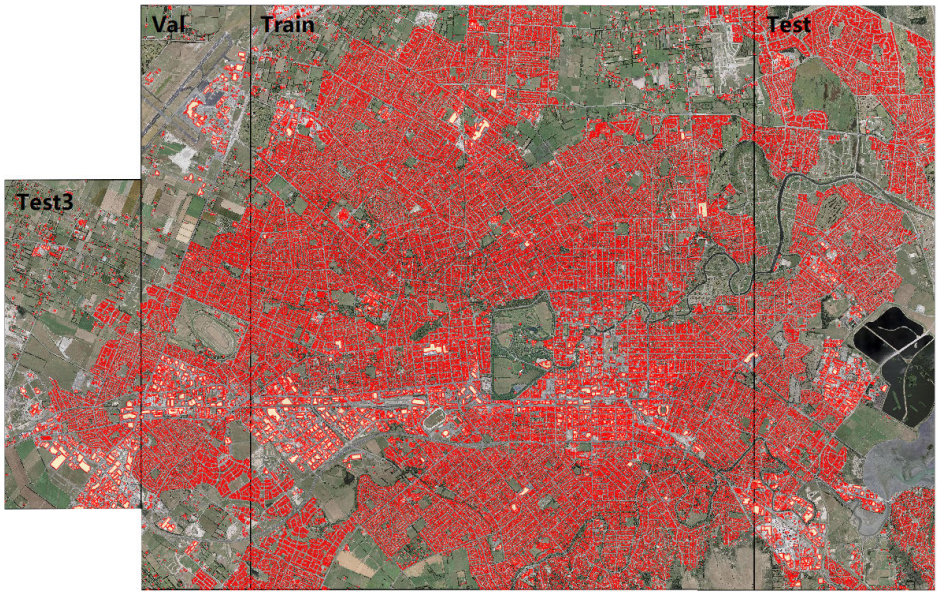


Fig. 6. Example images and the corresponding labels of the WHU dataset. The area in "Train" is used for training, the area in "Val" is used for validation, and the remaining in "Test and Test3" is used for testing.

1) Evaluation Method: The pixel classification evaluation algorithms were used to obtain the pixel-oriented building extraction results. P (precision), R (recall), ACC (accuracy), and $F1$ (harmonic mean) are the classical image evaluation algorithms. Precision and recall are a pair of contradictory measurement methods, and both can be used to measure the correctness of the building mask; the algorithms are as follows:

$$P = \frac{\mathrm{TP}}{\mathrm{TP} + \mathrm{FP}} \tag{3}$$

$$R = \frac{\mathrm{TP}}{\mathrm{TP} + \mathrm{FN}} \tag{4}$$

where TP is the true positive number, FP is the false positive number, and FN is the false negative number.

The accuracy and harmonic mean algorithms can been described as follows:

$$\mathrm{ACC} = \frac{\mathrm{TP} + \mathrm{TN}}{\mathrm{TP} + \mathrm{FP} + \mathrm{FN} + \mathrm{TN}} \tag{5}$$

where TP is the true positive number, FP is the false positive number, FN is the false negative number, and TN is the true negative number

$$F_1 = 2 \times \frac{P \times R}{P + R} \tag{6}$$

where P is the precision score and R is the recall score.

2) Footprints Distance Metric: Given the importance of building footprints in building boundary evaluation, we introduced a building footprint evaluation method known as the footprint distance metric. It accounts for the positional and shape differences by considering the building boundary as a sequence of connected boundaries instead of only point sets. In addition to the outstanding performance in evaluating the similarity of footprints, the footprint distance metric can address the case of an unequal number of footprints between each building.

The footprint distance metric determines the accuracy of the differences between the reference and synthesis shapes using the coordinate value of building footprints. The metric applies the following four steps for the differences between building A and B: 1) *point distance:* calculate the distance between the footprints in building boundary A and all the footprints in building B; 2) *edge distance:* calculate the minimum distance between the footprints in building A and all the boundary in building B, and then remove the boundary distance in which the vertical points are not in the building boundary; 3) *footprint distance:* from steps 1) and 2), each footprint in building boundary A is assigned a footprint distance metric by taking the minimum value from the point and edge distances; and 4) the footprint distance metric of buildings A to B is the average of the footprint distance. Details of the footprint distance metric are shown in Fig. 7; the orange and blue polygons indicate the building boundaries, the black solid line is the selected point distance or edge distance, the gray dashed line is the no-minimum point distance or edge distance, and the blue dashed line is the extended line of each building edge that is used to calculate the edge distance.

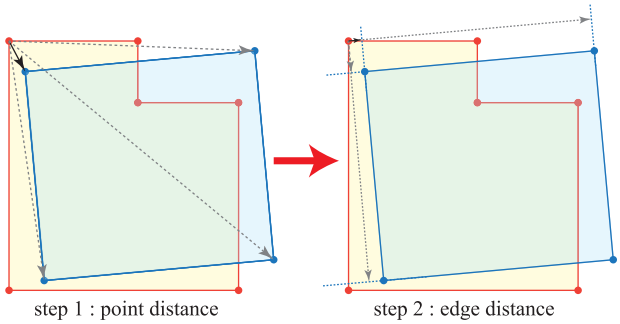


Fig. 7. Footprint distance metric in two building boundaries.

The footprints distance metric of building A to building B is as follows:

$$\tilde{p}(A, B) = \frac{1}{2m} \sum_{a_i \in A} \min_{b \in \partial B} \|a_i - b\| \tag{7}$$

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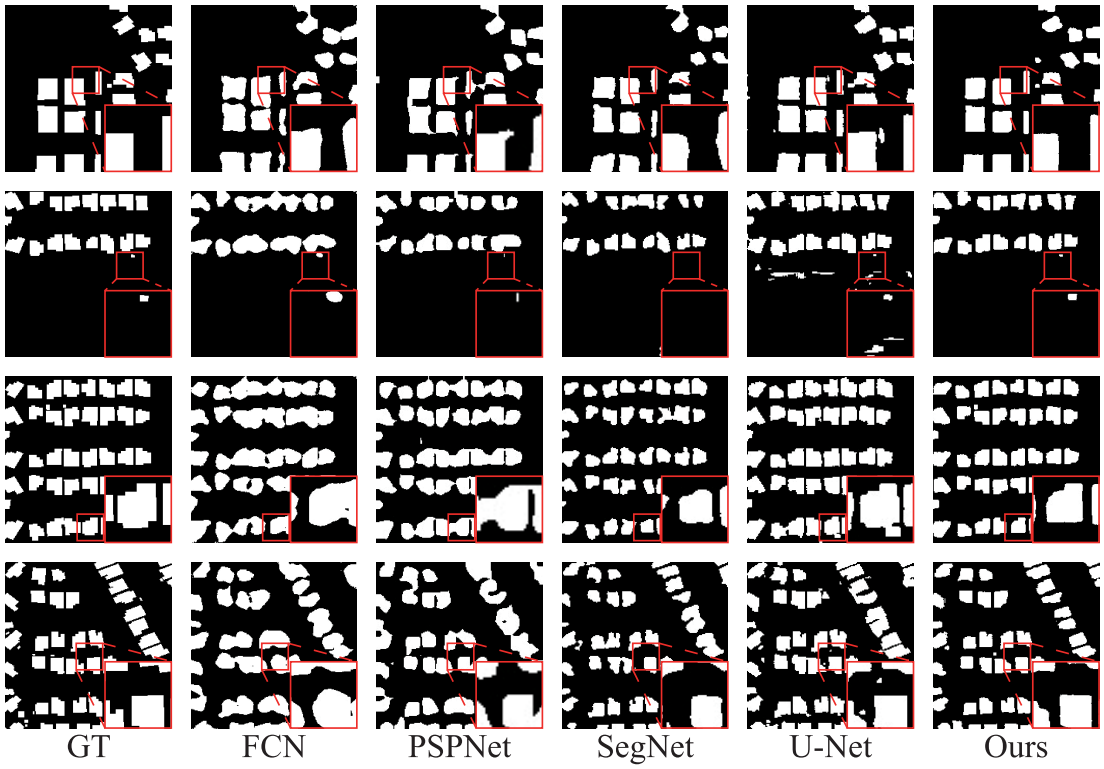


Fig. 8. Building boundary delineation results of the semantic segmentation methods applied on the Spacenet dataset.

where m is the footprints number of building A, a_i is a footprint in building A, b is the closest point, and ∂B is the point set of footprints and vertical points.

Similarly, $p(B, A)$ is the footprints distance metric of building B to building A, and the algorithm is as follows:

$$\tilde{p}(B, A) = \frac{1}{2n} \sum_{b_j \in B} \min_{a \in \partial A} \|b_j - a\| \tag{8}$$

where n is the footprints number of building B, b_j is a footprint in building B, b is the closest point, and ∂A is the point set of footprints and vertical points.

The difference in the footprints distance metric of each building boundary is as follows:

$$p(B, A) = \tilde{p}(A, B) + \tilde{p}(B, A). \tag{9}$$

C. Experiment

In this study, four state-of-the-art pixel-oriented methods, including FCN [36], SegNet [51], U-net, and pyramid scene parsing network (PSPNet) [52], were selected for comparison. These methods have been employed used in recent studies on building extraction. Besides traditional semantic segmentation, coarse-to-fine boundary refinement network (CBRNet) and high-resolution network (HRNet) are also introduced to evaluate the details of building boundaries. The CBRNet and HRNet are boundary feature enhancement methods used for the entire remote sensing image. This study used binary methods to compare semantic segmentation results that converted building coordinates to binary masks and took the average results. Moreover, to verify the efficiency of boundary refinement, the performance of the optimized boundary methods, namely, active contours via differentiable rendering network (ACDRNet) [18] and deep structured active contours (DSAC), was evaluated. ACDRNet and DSAC are building boundary point generation methods used in a single building object. This study introduced an excellent instance segmentation method [path aggregation network (PANet)] [53] to verify the effectiveness of the proposed method in the instance segmentation method for comparison. The quantitative evaluation and its results are presented in the figures and tables.[2]

1) Comparison of Semantic Segmentation Methods: Significant results have been obtained using the semantic segmentation method to extract building boundaries. The classical semantic segmentation algorithms—FCN, SegNet, U-net, and PSPNet—determined whether each pixel is a building object to detect building boundaries. These methods consider the high-dimensional semantic information of the building objects entirely, and the textural features can be the maximum reservation to identify building pixels. Figs. 8 and 9 show the extraction results obtained from this study and those from the semantic segmentation methods applied to the Spacenet and WHU datasets. The details of the building boundaries are shown in the magnified images below the actual results. The method used in this study can separate each building object, whereas the adhesion phenomenon occurred in semantic segmentation methods. It is well known that a remote sensing image has a complex scene in which another ground object always occludes the building boundary. From the building delineation results, it is clear that this study achieved an outstanding performance in complex remote sensing scenes where the occlusion of the boundary by trees occurs. In particular, this study achieved outstanding results in a building boundary

[2]The code of PANet: https://github.com/carlfu127/Keras-PANet.git

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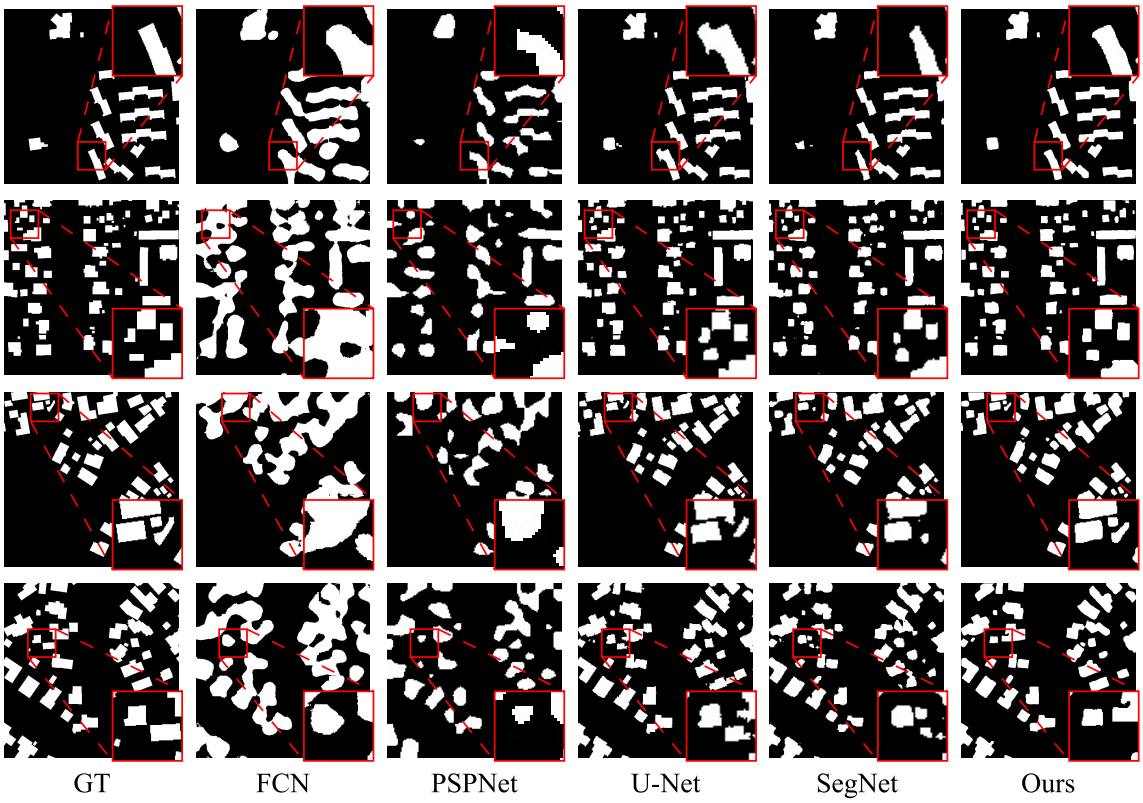


Fig. 9. Building boundary delineation results of the semantic segmentation methods applied on the WHU dataset.

TABLE I
EVALUATION SCORE OF THE SEMANTIC SEGMENTATION METHODS IN THE SPACENET DATASET

Methods	Recall	F1	Precision	Accuracy
FCN	0.8337	0.7088	0.6566	0.9220
SegNet	0.7497	0.6869	0.6845	0.9184
U-net	**0.8879**	0.7004	0.6247	0.9141
PSPNet	0.8146	0.6577	0.5984	0.8993
ours	0.8516	**0.7298**	**0.6942**	**0.9241**

TABLE II
EVALUATION SCORE OF THE SEMANTIC SEGMENTATION METHODS IN THE WHU DATASET

Methods	Recall	F1	Precision	Accuracy
FCN	0.6689	0.4832	0.6313	0.9143
SegNet	0.6644	0.5333	0.6794	0.9413
U-net	0.6792	0.5286	0.6804	0.9383
PSPNet	0.6170	0.4733	0.6555	0.9160
ours	**0.8531**	**0.7927**	**0.7517**	**0.9574**

with the regular shape of actual buildings, and other objects of cluttered similar buildings were ignored.

This study calculated these semantic segmentation methods' recall, $F1$, precision, and accuracy scores. The results of the Spacenet and WHU datasets are presented in Tables I and II. In the WHU dataset, this study achieved the best score on all evaluation metrics and showed outstanding performance in building boundary extraction. In the Spacenet dataset, it can be observed from the table that the model proposed in this study achieved the best $F1$, precision, and accuracy score. However, the recall score is lower than that of the U-net, which indicates that the U-net can find more buildings than our method. Because our method is an object-oriented method, it only preserves the building object with a high confidence level. Therefore, our method extracts a complete building object rather than more building objects, even if it is incomplete.

Except for the classic semantic segmentation methods, several semantic segmentation studies have examined the building boundary refinement problem. These methods proposed several optimized blocks to improve the imperfect edge information. CBRNet and HRNet focus on obtaining a perfect building outline. CBRNet uses a boundary refinement module that enhances the ability to perceive and refine building edges. It refines the process of building prediction by perceiving the direction of each pixel to the center of the nearest object to which it might belong. The HRNet uses a HRNet to maintain high-resolution representations through the entire process, which makes the resulting representation semantically richer and spatially more precise. The CBRNet and HRNet models were trained using the binary Spacenet and WHU datasets. The results are presented in Figs. 10 and 11 and Tables III and IV.

The enlarged frames in Figs. 10 and 11 list the details of the respective method. Our method accurately extracted the building boundary in the complex remote sensing scene, and the false retrieval and missed detection phenomena were reduced. Although the HRNet and CBRNet methods optimized the building boundary, our method obtained more refined details in the concave–convex area. This study solved the problem of misdetection observed in the two building optimized methods.

Table III shows that our model achieved the best recall, $F1$, and accuracy scores on the Spacenet dataset; however,

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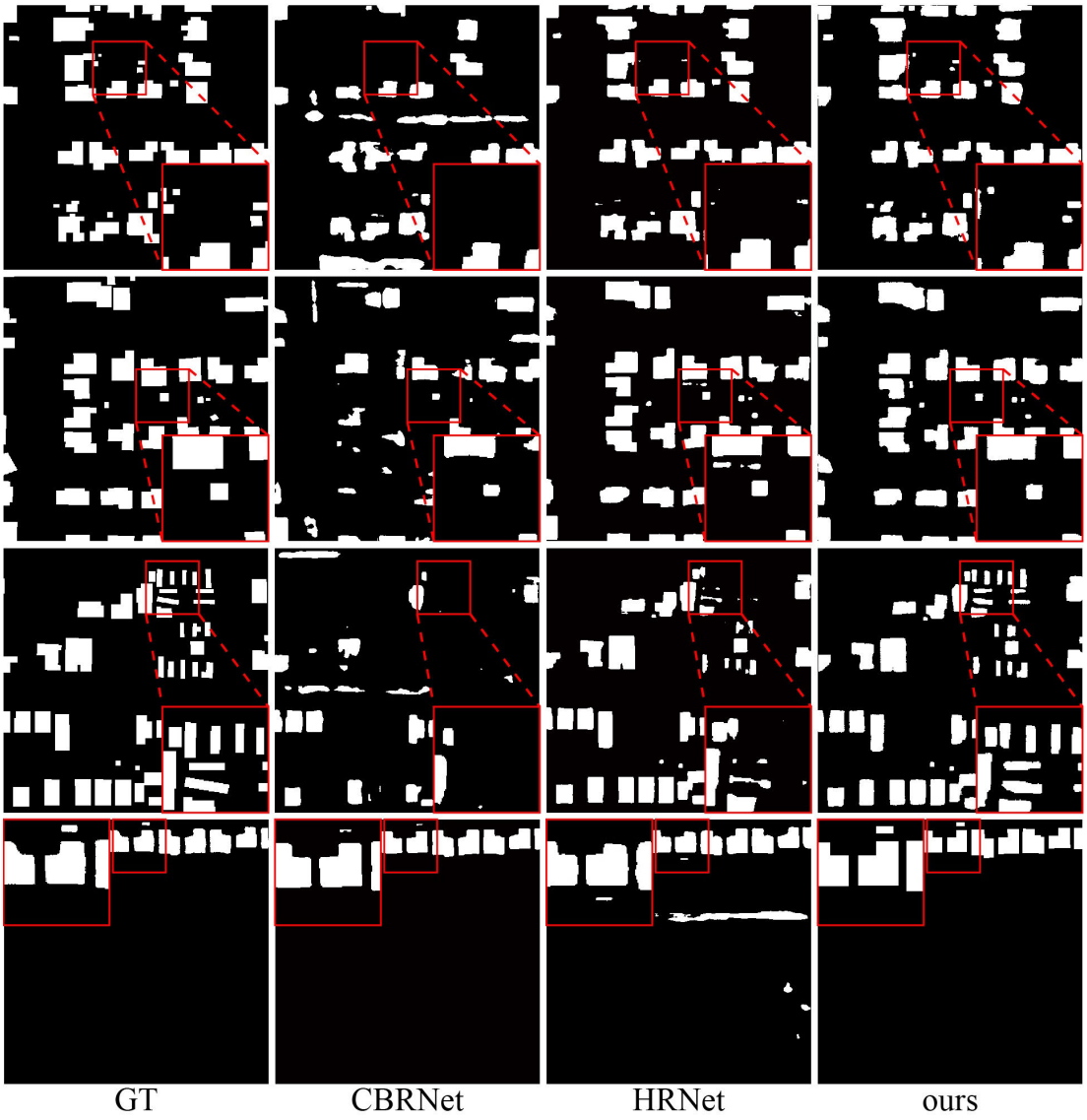


Fig. 10. Building extraction results of multiboundary optimized methods in the Spacenet dataset.

TABLE III
QUANTITATIVE EVALUATION SCORES OF THE MULTIBOUNDARY OPTIMIZED METHODS APPLIED ON THE SPACENET DATASET

Methods	Recall	F1	Precision	Accuracy
CBRNet	0.4839	0.4906	0.6581	0.8802
HRNet	0.7318	0.6782	**0.7201**	0.9181
Ours	**0.8516**	**0.7298**	0.6942	**0.9241**

TABLE IV
QUANTITATIVE EVALUATION SCORES OF THE MULTIBOUNDARY OPTIMIZED METHODS APPLIED ON THE WHU DATASET

Methods	Recall	F1	Precision	Accuracy
CBRNet	0.6717	0.6047	0.7334	0.9504
HRNet	0.6292	0.6351	**0.9294**	**0.9590**
Ours	**0.8531**	**0.7927**	0.7517	0.9574

its precision score is lower than that of HRNet. HRNet is an excellent boundary optimization method that provides superior building extraction results. Recall and precision are contradictory indicators; our method achieved a recall of 0.8516, which is 0.12 higher than that of HRNet. However, the precision and accuracy scores were lower than those of HRNet on the WHU dataset. Our methods achieved the best recall and $F1$ score. The $F1$ score is used to measure the quality of the model. While validating the HRNet and CBRNet methods, we found that the results without building tags in the test dataset were also used to derive indicators. However, this method discards these data, resulting in a decline in evaluation indicators.

2) Comparison With the Automated Building Labeling Method: Recent studies in automatic annotation used a different approach for optimization and employed a neural renderer to create the polygon from its vertices. In the method, the input of the automated building labeling method is the single building patch, and these methods focus on building edge optimization. To evaluate the performance of building edge optimization, we compared the extracted building boundary results obtained using ACDRNet and DSAC. The two methods

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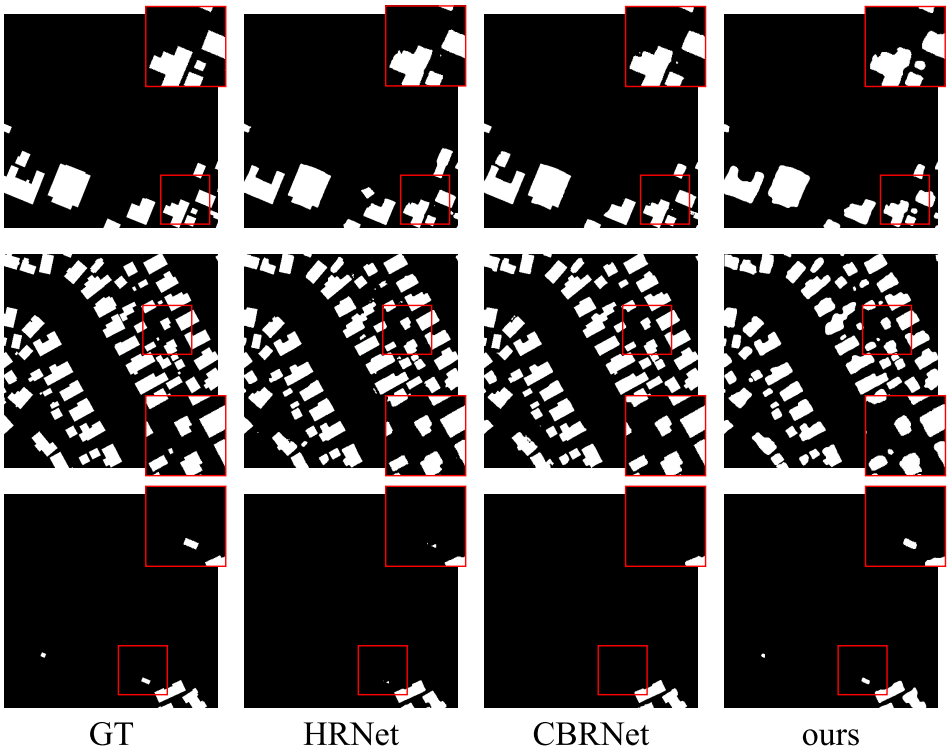


Fig. 11. Building extraction results of multiboundary optimized methods in the WHU dataset.

TABLE V
QUANTITATIVE EVALUATION RESULTS OF SINGLE BUILDING BOUNDARY OPTIMIZED METHODS APPLIED ON THE SPACENET DATASET

Methods	Recall	F1	Precision	Accuracy
ACDRNet	0.8764	0.7997	0.7660	**0.9804**
DSAC	0.7678	0.6249	0.6345	0.9692
Ours	**0.9022**	**0.8644**	**0.8388**	0.9425

combine CNN models with the classical polygon-based active contour model and focus on the minimum energy of the building edge. Because these methods can extract only a single building boundary, this method cuts the original training datasets into 256 × 256. It kept the extracted building in the center of the training data. Therefore, the results of the methods contain considerable background area, which is the supplementary area and can maintain the center position of marginal buildings. A summary of the qualitative assessment results is shown in Fig. 12.

Although ACDRNet and DSAC have been widely employed in recent remote sensing images, the disadvantage of these methods is that they can optimize only one building. In this experiment, the results were calculated using 1000 buildings. Therefore, the quantitative evaluation score differs from that obtained in other experiments. Using a fixed-size image in training requires multiple background pixels, which are easy to identify and significantly increase the accuracy score. The quantitative evaluation results of ACDRNet and DSAC are presented in Table V. Our method outperformed ACDRNet and DSAC on most metrics.

3) Comparison With the Instance Segmentation Method: To evaluate the performance of the instance segmentation, we compared the extracted building boundary results with PANet. PANet is a classic instance segmentation method; it enhances the feature hierarchy with accurate localization signals in lower layers by bottom-up path augmentation and adaptive feature pooling. This method links the feature grid, and all feature levels to make helpful information in each feature level propagate directly to the following proposal subnetworks. Moreover, this method adds the number of parameters, the frames/s (FPS), and the per batch time. The summary of the qualitative assessment results is shown in Fig. 13.

As shown in Fig. 13, our method provided superior boundary results in that the corners of the building have a more precise contour, and sawtooth boundaries have also been considerably improved. In addition, compared with other object-oriented building extraction methods, the phenomenon of building error detection and missing detection in this method has been significantly improved. The quantitative evaluation results of these methods are presented in Table VI, and the efficiency evaluation results are shown in Table VII. Similar to this article, the code of PANet is revised from the original Mask R-CNN, which has the same data structure and code logic. Our method achieved the best scores on the evaluation indicators compared with the instance segmentation methods. Although PANet has achieved high values in average precision (AP) metrics, the proposed method is far superior to PANet in terms of the average recall (AR) and the footprint distance metrics. This method has improved the

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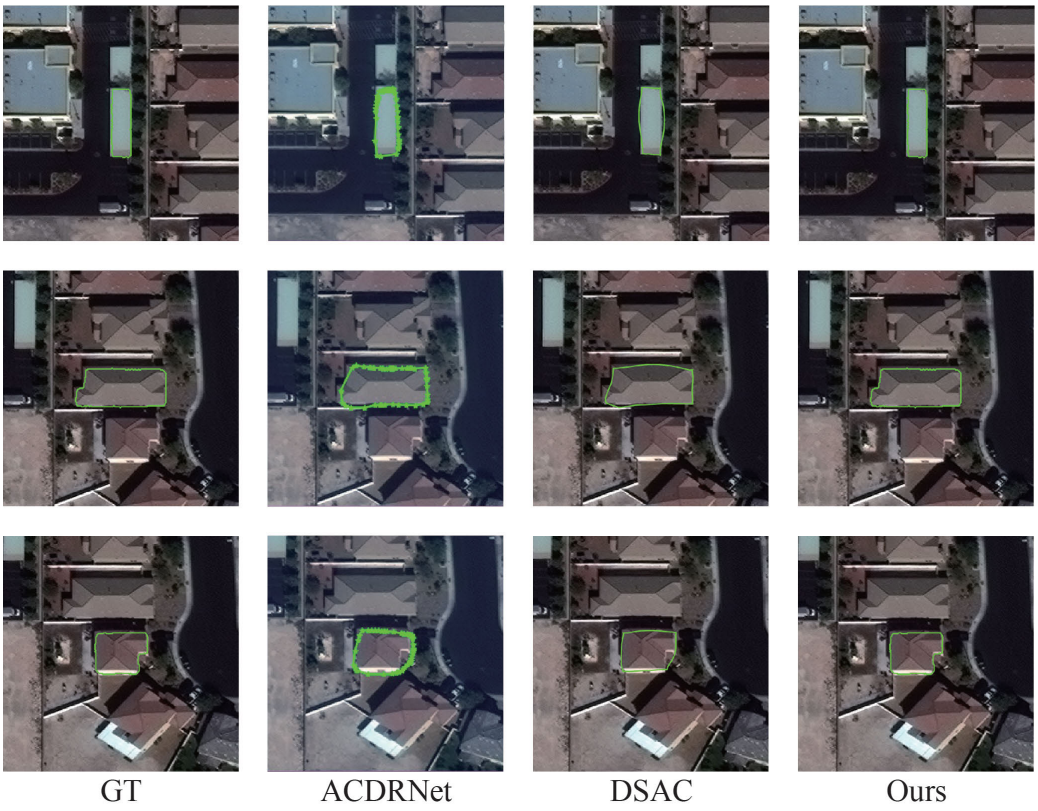


Fig. 12. Comparison of the results for some buildings. From left to right, the following hold: 1) ground truth; 2) ACDRNet; 3) DSAC; and 4) the proposed method.

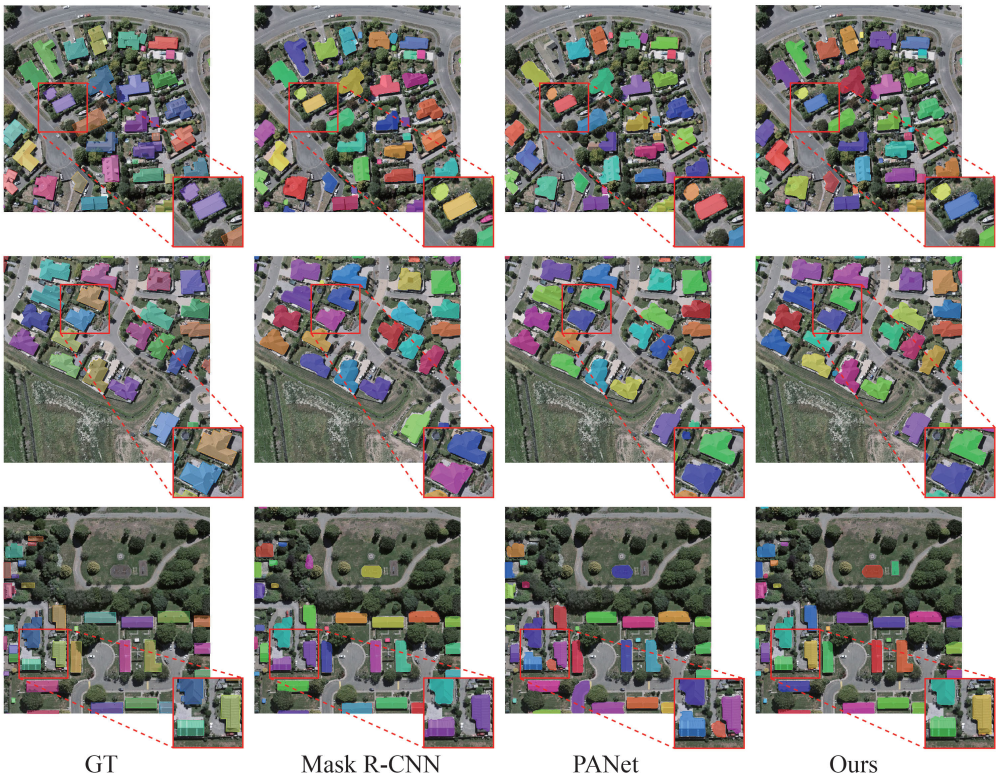


Fig. 13. Building extraction results of the instance segmentation methods applied on the WHU dataset.

ability of Mask R-CNN to extract building edge information by optimizing the network model, and it performs better in terms of efficiency than the excellent example segmentation method PANet.

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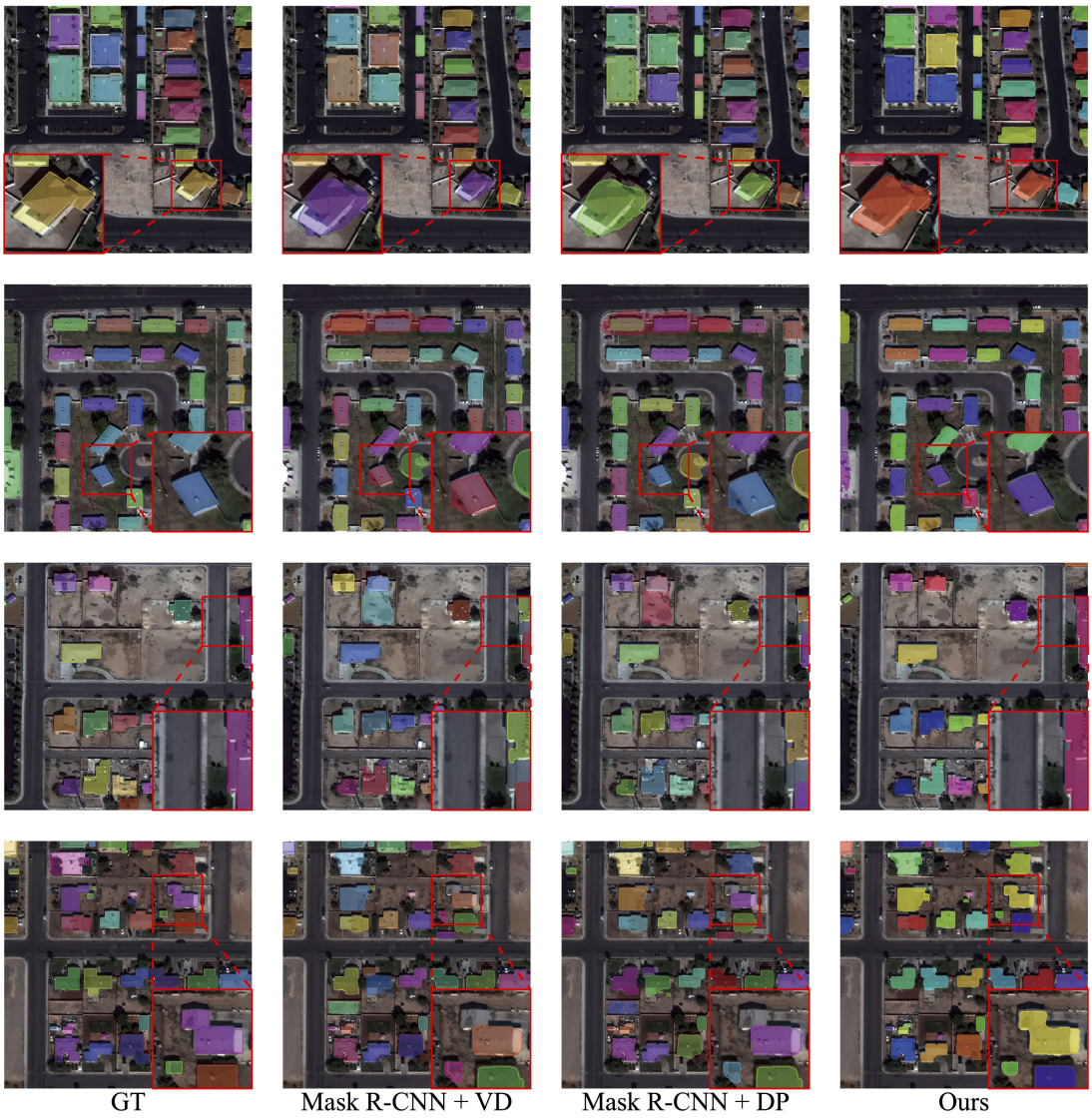


Fig. 14. Results of the simplification methods applied to the Spacenet dataset. From top to bottom, the following hold: 1) ground truth; 2) Mask R-CNN + DP; 3) Mask R-CNN + VD; and 4) the proposed method.

TABLE VI
QUANTITATIVE EVALUATION RESULTS OF THE INSTANCE SEGMENTATION METHODS

Methods	AP	AR	The footprint's distance metric
Mask R-CNN	0.9187	0.8878	1.9778
PANet	**0.9293**	0.8705	2.2872
Ours	0.9285	**0.9009**	**1.6807**

TABLE VII
NUMBER OF PARAMETERS AND COMPUTATIONAL EFFICIENCY

Methods	Parameters	Per-batch time	FPS
Mask R-CNN	**40,390,686**	**0.27155132**	**11.60763249**
PANet	41,756,334	2.463201871	0.339167711
Ours	41,430,972	0.286030397	11.75419178

V. DISCUSSION

A. Simplification for Instance Segmentation Contour Results

The building boundary results extracted using image segmentation and deep learning methods are constructed using several broken lines. The results do not need to conform to the shape of the building footprints. To verify the performance of this method in the end-to-end strategy, we calculated the simplification results of the Mask R-CNN method using Douglas–Peukcer (DP) [54] and vertical distance (VD), a classical polygon simplification method widely used in the field of remote sensing. DP and VD were used to simplify the Mask R-CNN results, which can verify the validity of our method. In this experiment, the threshold for these simplification methods was 0.3. The building boundary extraction results are shown in Fig. 14. We obtained more accurate results in the complex building boundary shape, and the footprints were closer to the edge of the building inflection point. This method obtained has fewer vertices results in case extra prior parameters are unnecessary.

The qualitative evaluation results of this study highlight further advantages of the proposed method. Regarding the distance evaluation index, the lower the footprint distance metric score, the better the building boundary obtained. Table VIII demonstrates that this method performed better regarding

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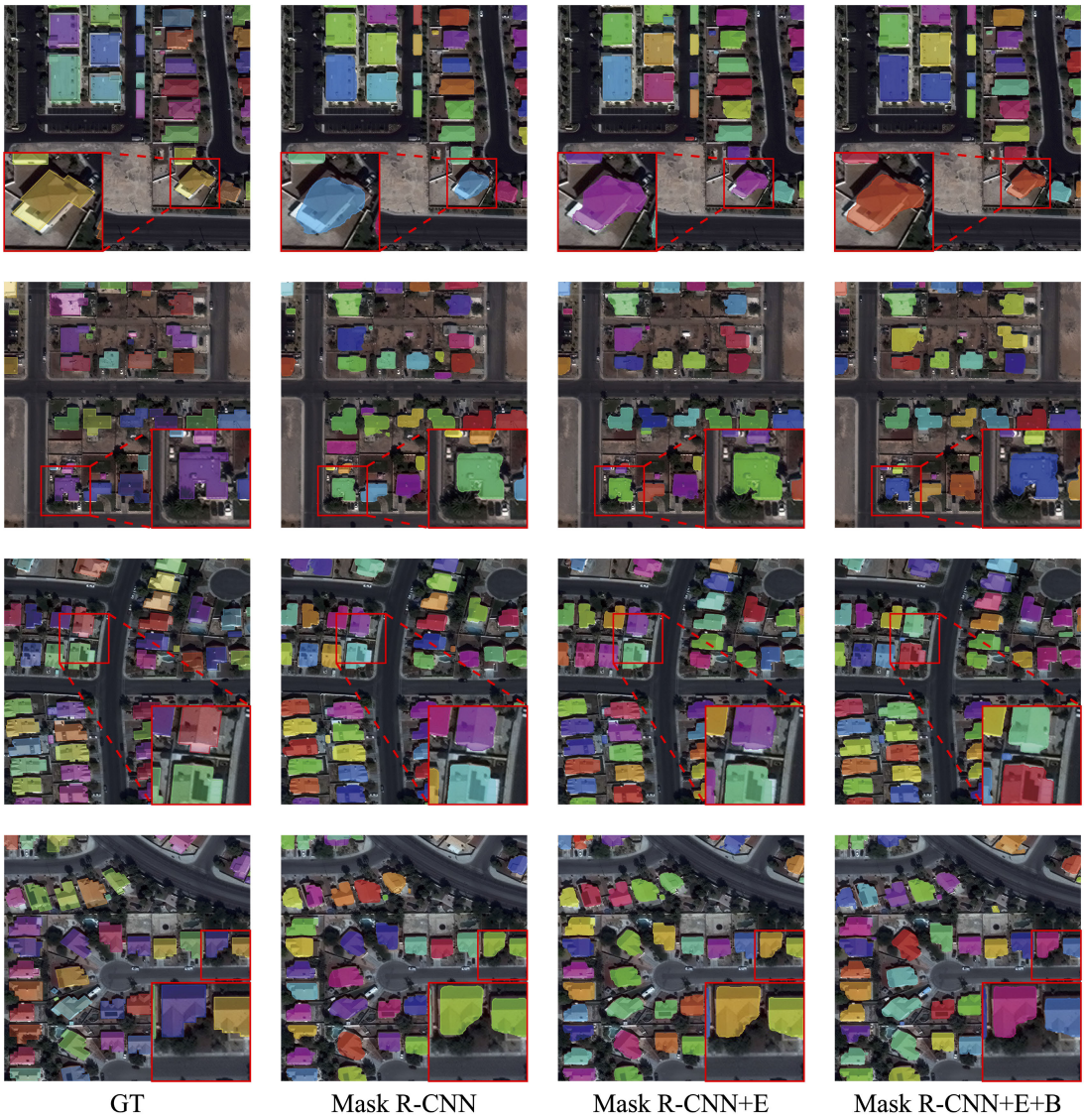


Fig. 15. Extracted building results of the ablation experiment conducted on the Spacenet dataset.

TABLE VIII
EVALUATION SCORE OF THE SIMPLIFICATION METHODS APPLIED ON THE SPACENET DATASET

Methods	AP	AR	The footprint's distance metric	Number
Mask+DP	0.9165	0.8434	3.7295	1433
Mask+VD	0.9231	0.8261	3.8966	1441
Ours	**0.9242**	**0.8628**	**2.9864**	**1031**

the AP, AR, and footprint distance metrics. In this table, "Mask + DP" is the simplified Mask R-CNN result by DP, and "Mask + VD" is the simplified Mask R-CNN result by VD. In particular, the minimum footprint distance metric score indicates that the footprints of our results are closer to the ground-truth building boundary. The number of unmatched buildings while calculating the footprint distance metric score is presented in the last column; our method had the least number of unmatched buildings, indicating that it yielded minimal error objects.

B. Ablation Experiment

This study designed a boundary shape-preserving model to extract the details of the building boundary better. First, the super-resolution block was designed to combine Mask R-CNN. The designed shape-preserving network further refined the indeterminate boundary points and solved the problems occurring in the concave–convex areas of buildings. In addition, the regular boundary network was designed to improve the sawtooth boundary, which generates boundary information, and was optimized using the boundary loss. The extracted results of the ablation experiment are shown in Figs. 15 and 16 and Tables IX and X. "E" represents the edge point enlargement algorithm, "B" denotes the boundary loss, and the "Mask R-CNN + E + B" indicates this study.

Unlike the original Mask R-CNN model, this study thoroughly considered the boundary defect in extracting building boundaries. As shown in Figs. 15 and 16, the sawtooth boundary showed significant improvement. The quantitative evaluation results are presented in Tables IX and X.

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Fig. 16. Extracted building results of the ablation experiment conducted on the WHU dataset.

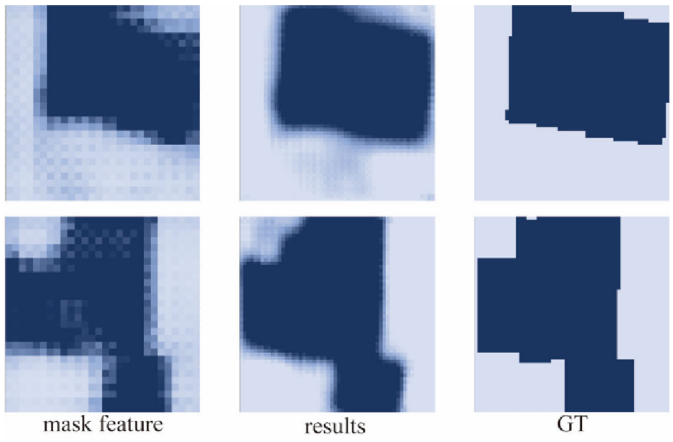


Fig. 17. Visual results of SPM.

TABLE IX
QUANTITATIVE EVALUATION RESULTS OF THE ABLATION EXPERIMENT IN THE SPACENET DATASET

Methods	AP	AR	the footprint's distance metric
Mask R-CNN	0.9164	0.8448	3.6295
Mask R-CNN+E	0.9225	0.8582	3.0404
Mask R-CNN+E+B	**0.9242**	**0.8628**	**2.9864**

TABLE X
QUANTITATIVE EVALUATION RESULTS OF THE ABLATION EXPERIMENT IN THE WHU DATASET

Methods	AP	AR	the footprint's distance metric
Mask R-CNN	0.9187	0.8878	1.9778
Mask R-CNN+E	0.9182	0.8895	1.9397
Mask R-CNN+E+B	**0.9285**	**0.9009**	**1.6807**

In particular, the significant improvement in the footprint distance metric indicates that the results obtained in this study are more geometrically similar.

This method proposed a SPM to solve the problem of building shape representation. To further validate the SPM's ability to express the shape of buildings, we presented the input and output visualization results in Fig. 17. The input of SPM is the fused feature by coarse- and fine-grained features. The final building extraction result is the binary building.

Where the mask feature is coarse-grained information, the result is the feature representation results. Own to the benefit of the upsample process, the uncertain edges within the building have been improved, and the boundary has also been more clearly outlined.

VI. CONCLUSION

Building extraction methods have recently focused on enhancing regional accuracy without obtaining refined building boundaries. The concave–convex area, sawtooth boundary, and insensitive building evaluation method problem limit the accuracy of building extraction. We proposed a boundary-enhanced building extraction method for visible

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light remote sensing images. The following improvements were made to the Mask R-CNN model to realize outstanding building boundary extraction: 1) the improved Mask R-CNN incorporates an edge point enlargement algorithm to optimize the mask branch, and a boundary loss function is proposed to evaluate the edge and footprint errors and 2) this study used the footprint distance metric to calculate the footprint distance between each mask. It was enhanced using the extra distance of the polygon center point to identify each boundary in the same building. Our method accurately delineated building boundaries for the open-source benchmark datasets, Spacenet and WHU datasets. The end-to-end building boundary delineation method performed well in complex remote sensing scenes. The building extraction problem will be further investigated in the future, focusing on the topological errors (such as the unclosed building boundary, the gap in the building boundary, and polygon overlap) in the extracted footprints.

ACKNOWLEDGMENT

The authors wish to thank the anonymous reviewers and editors for their detailed comments and suggestions.

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